

James LaFemina

Education

Ph.D. Student, Dept. of Biomolecular Engineering

University of California, Santa Cruz 2023 – Present

B.S. Biochemistry & Molecular Biology, Highest Honors in the Major

University of California, Santa Cruz (2023)

Research Experience

Sharf Lab, Dept. of Biomolecular Engineering, UC Santa Cruz

Interdisciplinary mission to decode how human brain circuits emerge, interact, and transmit information by combining 3D stem cell-derived brain models with high-resolution neuro-electronics to map activity at single-cell resolution across functional networks.

Graduate Researcher | Sept 2024 – Present

My research aims to uncover the early drivers of Parkinson's disease (PD) using genomics and systems approaches to link molecular mechanisms, gene regulatory dynamics, and circuit-level activity in human-derived brain models—ultimately enabling discovery of disease-modifying interventions.

Selected Achievements:

- Optimized midbrain organoid differentiation protocols (including neurotoxic lesioning) to develop late-stage PD *in vitro* models.
- Developed and executed immunohistochemistry workflows to visualize and quantify dopaminergic neuron depletion in organoids.
- Performed high-density microelectrode array (HD-MEA) experiments on fused midbrain-thalamic organoids (assembloids).
- Constructed a custom data analysis pipeline using Python programming and statistical methods to characterize functional network breakdown following neurotoxin treatment.
- Collaborated with fabrication, electrophysiology, and computational teams within the Braingeneers consortium to iteratively refine experimental design, data-analysis approaches, and interpretation of results.

Boeger Lab, Dept. of Molecular, Cell, and Developmental Biology, UC Santa Cruz

Research aimed at understanding how stochastic transcriptional bursts and ATP-dependent chromatin remodeling together shape gene regulation specificity, using a combination of theoretical modeling and single-molecule analyses to reveal how dynamic chromatin states resolve molecular noise into precise cellular outcomes.

Graduate Researcher | June 2023 – Sept 2024

I investigated how disruptions in histone acetylation and nucleosome dynamics destabilize chromatin architecture and drive aneuploidy in budding yeast. To dissect this relationship, I used reverse genetics, chromatin biochemistry, quantitative single-cell analysis, and statistical modeling to link molecular modifications at the nucleosome level to large-scale genomic instability.

Selected Achievements:

- Designed experiments to test whether improper H3K56 acetylation induces re-replication rather than chromosomal missegregation as the primary driver of aneuploidy.
- Developed mutant yeast strains mimicking constitutive H3K56 hypo- and hyperacetylation to test their impact on genome stability.
- Employed a fluorescent reporter system to infer chromosomal copy number variations via flow cytometry, enabling real-time tracking of aneuploidy in single cells.
- Validated this mechanism by adapting and modernizing the Luria-Delbrück fluctuation test to quantify mutation rates under different acetylation states.
- Applied topological assays using DNA supercoiling and topoisomerases to quantify nucleosome occupancy under various genetic perturbations.
- Used MNase digestion and nanoNOME (single-molecule chromatin accessibility and methylation sequencing) to analyze chromatin structure and accessibility.
- Engineered inducible anchor-away systems to acutely deplete key regulatory proteins (e.g., TBP) and assess nucleosome dynamics in real time

Undergraduate Researcher | March 2023 – June 2023

Senior Thesis: *ASF1-Mediated Histone H3K56 Regulation in Yeast Chromatin Stability*

Advisor: Professor Hinrich Boeger

Teaching and Mentorships

Teaching Assistant

University of California, Santa Cruz

- *Biochemistry III* (Spring 2023 and Spring 2024)
- *Eukaryotic Molecular Biology* (Winter 2024)

Undergraduate Mentorships

- Mentored a total of 6 undergraduates at my bench in the Boeger Lab.

Professional Summary (Prior Career)

Sales executive and leadership development specialist with a 20-year track record of helping senior leaders at some of the world's top brands solve complex problems with business intelligence, software and analytics tools, and experiential learning programs.

Selected Achievements:

- Led enterprise sales of high-impact experiential leadership programs in partnership with NASA targeting senior leadership teams and high-potential cohorts at Fortune 500s.
- Collaborated directly with executives to shape learning engagements that addressed pressing organizational challenges and promoted adaptive team leadership..
- Built and managed pipelines as both an internal consultant and external on-demand VP, helping startups and Fortune 500 clients alike solve high-stakes leadership and performance challenges.